

Smart Home AIoT – Test Data Collection Sheet – Phase I – Integration Tests

Senior Project CE08 – Bangkok University

Generated: 2026-05-25 22:46

Instructions for use:

1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:

- Pass Rate = $(\text{Pass Count} / 10) \times 100\%$
- Avg RT = $(\text{response times}) / 10$
- Variance = $(x_i -)^2 / n$
- Std Dev = $\sqrt{\text{variance}}$

Coverage Matrix

Phase	Description	Count
Phase I	Phase I – Integration Tests	12
	TOTAL	12

Phase I – Integration Tests

I01	HA Dashboard Latency				Phase I		
Method: click switch on Lovelace, measure ms							
Expected: < 500 ms							
#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10		Pass Rate: ____%		Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: voice + measure

Expected: 1-3 seconds

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: LINE + measure

Expected: 2-5 seconds (Claude API)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%		Avg RT: ____ ms		Std Dev: ____ ms
Comments:							

Method: Sleep + walk PIR

Expected: ไฟไม่เปิด (adaptive OFF)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: Sleep + PIR bedroom

Expected: Tuya dim 10% warm

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ___/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

I06	Sleep path light auto-off 2 min	Phase I
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Method: after path light

Expected: auto OFF after 2 min

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10 Pass Rate: ____% Avg RT: ____ ms Std Dev: ____ ms

Comments:

Method: HomeSleepHomeSleepAway (5s)

Expected: final state correct, no stuck

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

I08

Away during adaptive

Phase I

Method: Set Away while adaptive light ON

Expected: ไฟปิด, lock ทำงาน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: Set Sleep while Relax fan ON

Expected: fan OFF, curtain close

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

I10

ESP32 reboot recovery

Phase I

Method: unplug ESP32 #1 10s + plug

Expected: reconnect MQTT + retained state restore

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: restart HA container

Expected: all entities back online

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

I12

WiFi disconnect/reconnect

Phase I

Method: disconnect 30s + reconnect

Expected: webhook auto-heal

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments: